Paul Abellana IT342 – G4 3/11/202 Phase 1 Development – Google Sign-In Authentication Integration

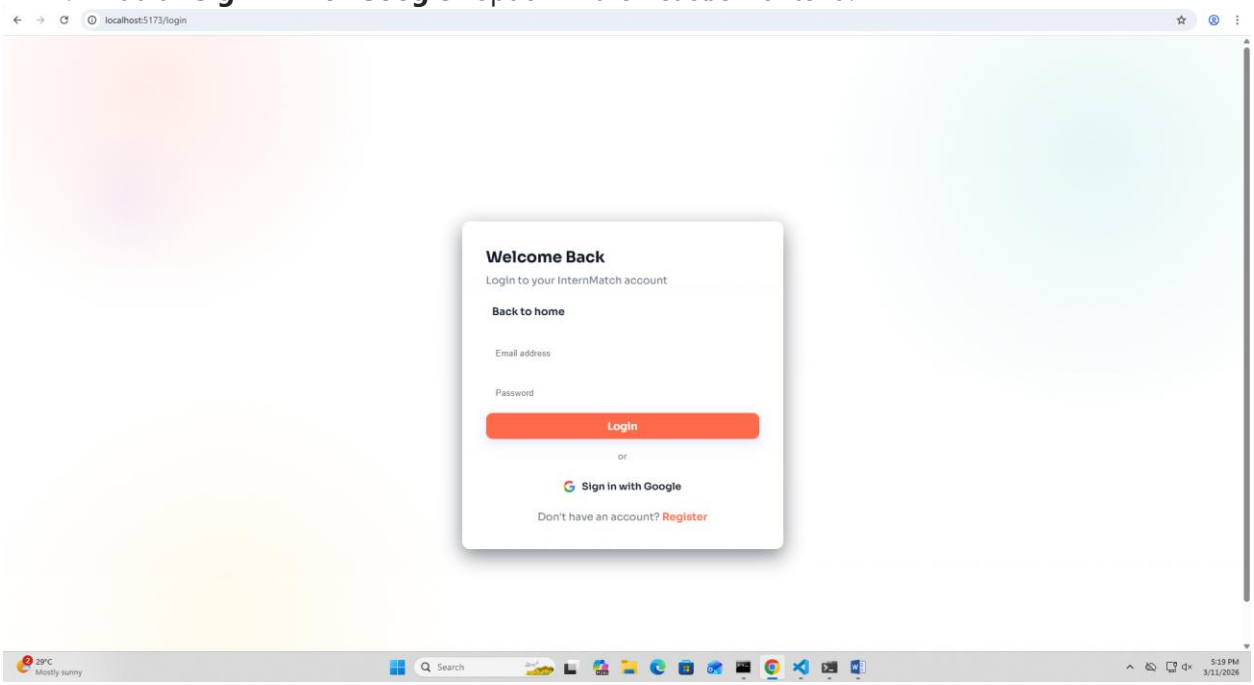
1. **Implementation Summary**

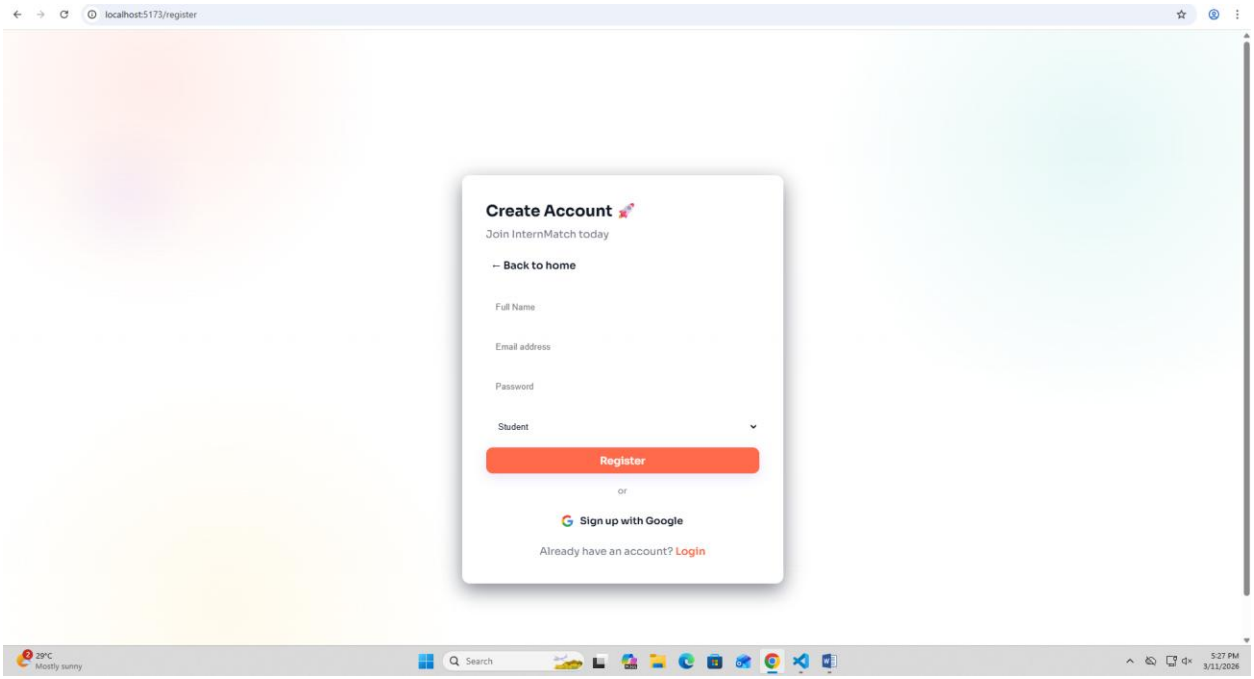
- InternMatch
- Brief description of how Google Login was implemented
  - Google login was integrated in the front end using a temporary direct OAuth redirect flow.
  - The app redirects users to google account selection, then receives OAuth tokens on the frontend callback URL
  - The call back is handled in Home.jsx where token/user info is parsed and mapped into the app's auth state using loginWithOAuth
- Backend technologies/libraries used
  - Springboot 3 for backend framework
  - Spring Security for authentication/security config
  - spring-boot-starter-oauth2-client for OAuth client support
  - JJWT for JWT generation/validation
  - Spring Data JPA + MySQL for persistence
  - Lombok for reducing boilerplate
  - Maven for build/dependency management
- Frontend integration summary
  - React + Vite frontend includes Google buttons in both Login and Register screens.
  - Login/Register pages initiate Google OAuth redirect
  - Callback handling in Home.jsx parses OAuth response and signs the user into app context
  - Auth state is persisted in local storage
  - Role based navigation redirects authenticated users to the correct dashboard (student,employer,admin)

Github link: https://github.com/0xSendo/IT342_G4_Abellana_Lab1

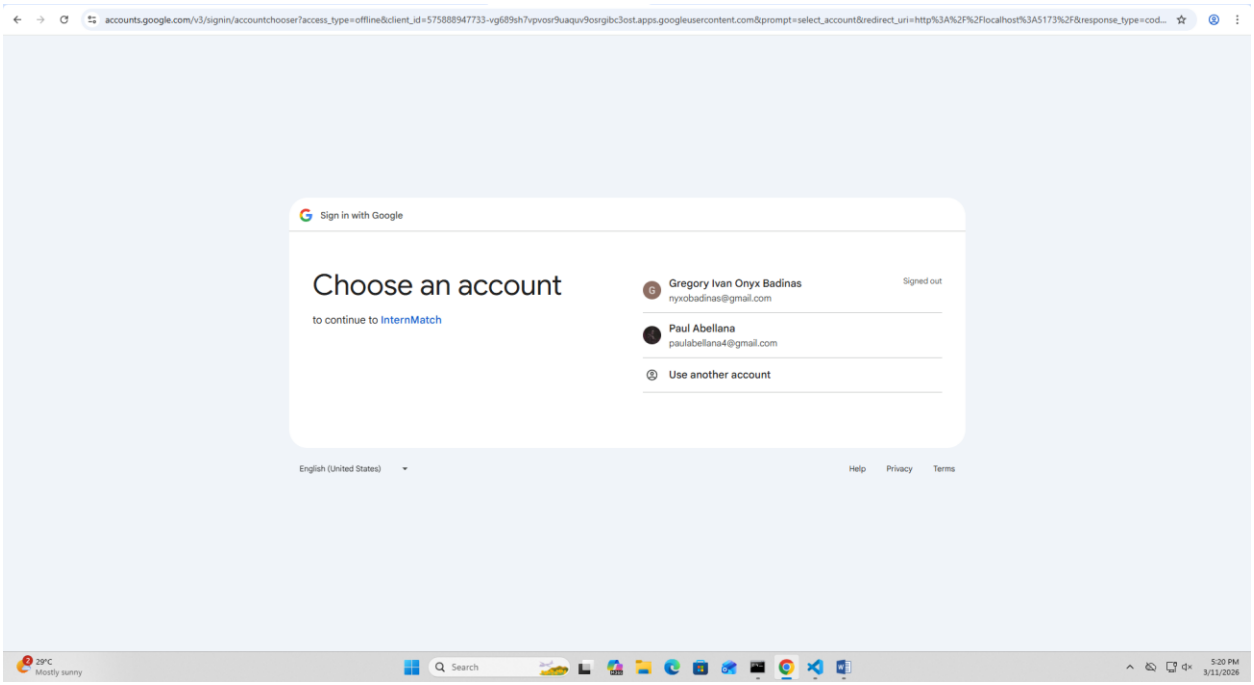
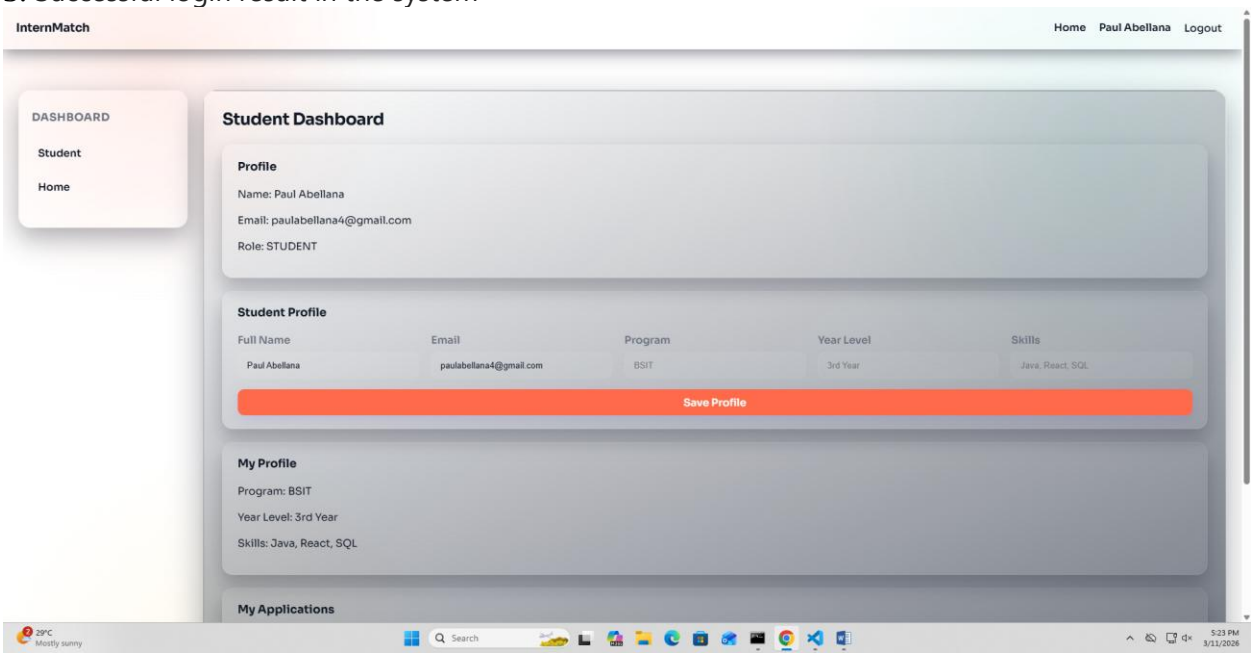
2. **Proof of Implementation**

1. Add a **"Sign in with Google"** option in the ReactJS frontend.





- 2. Google account selection or authentication page



- 3. Successful login result in the system